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CRITICAL MITIGATION ELEMENT (CME) METHODOLOGY
GUIDELINE (INCLUDING CONTROL SYSTEMS CME)

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1.0 PURPOSE

The general objectives of the Idd El Shargi Risk Management Guidelines (RMG) within the Idd El Shargi RM process are to:

- Allow a clearer definition of requirements with regard to the Idd El Shargi RM program and related policies and procedures;
- Allow specific topics to be addressed in detail relevant to the application of the RM element, including examples where relevant, and
- Act as a basis for further information, understanding and training.

Note: These documents should be considered as guidelines only and not mandatory unless otherwise stated, the use of the words, 'shall' and 'should' will indicate a mandatory requirement. For further clarification on content and intent contact Idd El Shargi HSE Risk Engineer.

The purpose of this Guideline is to provide guidance on when and how to apply CMEs when calculating Consequences associated with an Event Scenario; and to ensure that, when evaluating Risk, a consistent approach to the application of Critical Mitigation Elements is applied.

2.0 SCOPE AND APPLICABILITY

This Guideline describes Critical Mitigating Elements (CME) in two parts. The first part of the Guideline concerns the general application of CMEs in Process Hazard Reviews and Risk Evaluation. The second part of the Guideline discusses how CMEs specifically apply to Control Systems.

The Guideline specifically applies to all personnel within Idd El Shargi who are required to carry out Process Hazard Reviews, using the, Risk Evaluation and Response Procedure (IS-HSE-PRC-301), and the HSE Process Hazard Review (PHR) Procedure (IS-HSE-PRC-308).

CMEs are to be used for calculating Event Scenario probabilities in the Idd El Shargi Risk Management program except when:

- Event Scenarios are more appropriately evaluated based on Scenario-Based Analysis.
- More rigorous methods are necessary to refine probability estimates by Quantitative Analysis methods.

This Guideline applies to all personnel within Idd El Shargi plants and facilities (PS-1, ISND, ISSD and ISHT), who are involved in Risk Management, including Employees, contractors, and subcontractors, whether engaged directly by QatarEnergy or by others, who will perform work or services of any kind at any Idd El Shargi facilities and/or locations.

3.0 GUIDELINE

3.1 CRITICAL MITIGATION ELEMENT (CME)

A CME is an independent and reliable device or system that is capable of:

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- Preventing initiation of an Event Scenario,
- Preventing loss of Hazardous Material containment, or
- Substantially mitigating one of an Event Scenario's Consequences. Failure of one CME should not prevent another CME from performing its function. CME should be maintained to the reliability level assumed in the risk analysis.

CMEs are utilized in determination of Risk Levels when studying Event Scenarios.

3.1.1 CME Description

A Critical Mitigation Element will be:

- Reliable in preventing the initiation of the Event Scenario (e.g., where a spurious full- open failure of a relief valve is the initiating event, a CME is credited when there is a very low Probability of the relief valve failing in this way (e.g. the valve is operating in clean service) and all;
- Effective in preventing or significantly mitigating the Event Scenario Consequences when it functions as designed;
- Independent of the Initiating/Escalating Event(s), meaning that initiating and escalating events have not reduced the effectiveness of this protective feature and there are no components common with other CMEs already credited for the same Event Scenario (e.g., two ESDs controlled by one SIL1 PLC = 1 CME). Similarly, if a controller operating within a particular control system such as a Distributed Control System (DCS) or Programmable Logic Controller (PLC), is identified as a cause of the Event Scenario, no more than one CME can be claimed from the same control system. For example, if a Level Indicator Controller fails leading to potential high level and overpressure, only one CME can be claimed for a Pressure Indicator Controller and a Level Switch High, either of which may prevent the scenario, if they operate within the common control system, whereas two CMEs could be credited if they operate in different control systems.
- Documented (i.e., listed in PHR or Risk Evaluation) to validate assumed effectiveness in terms of Event Scenario Prevention or Mitigation.

3.1.2 CME Effectiveness Evaluation

The effectiveness of a CME may be evaluated using the following criteria:

- Can the device or system be relied upon to not fail in a way that would initiate an Event Scenario and, therefore, prevent initiation of the Event Scenario?
- For active CME devices or systems, can the device or system detect the condition that requires it to act (e.g., a process variable or an alarm) to prevent or mitigate the undesired Event Scenario Consequences? If the device or system CME cannot detect the condition and generate a specific action, then the device or system is not a CME;
- Is there sufficient time for the device or system to detect the condition, process the information, make a decision and take the required action, and for the required action to take effect to prevent or mitigate the undesired Event Scenario Consequences? If time is not sufficient to prevent or mitigate the

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undesired Event Scenario Consequences, then the device or system is not a CME;

- For active CME devices or systems, does the device or system have adequate capacity to take the required actions in the time available (e.g., relief valve orifice size or dike volume) to prevent or mitigate the undesired Event Scenario Consequences? If the device or system does not have the capacity to take required actions, then the device or system is not a CME;
- Is the strength of the device or system adequate for the required action (e.g., physical strength of blast wall or dike; ability of valve components to operate under the conditions that would be present for a particular scenario; or human strength within physical capabilities for the required action) to prevent or mitigate the undesired Event Scenario Consequences? If the device or system does not have the strength to take the required action, then the device or system is not a CME; and,
- Are there any Conditioning Factors that may impair the effectiveness of a CME (e.g. temporary repairs/wraps, documented failures, NROs (non-routine operations), LTIs (long term inhibits), Outstanding actions from related PHAs, PSSRs and Punch lists.

3.1.3 CME Application

The tables below in Appendix 1 provide a listing of CMEs grouped by those that prevent a hydrocarbon and/or toxic release and by those that prevent each final Consequence category. Each CME is given a reference name for ease of reference, and their use is mandatory. Note that a CME credit can only be taken once – not in each CME Group, unless there is a totally independent system that accomplishes that mitigation.

3.1.4 Key Mitigation Elements (KMEs) That Support CMEs

CMEs are robust mitigation measures that significantly reduce the probability of a consequence, their integrity must be maintained in order to achieve the intended Risk Level. Further, if the impact of supporting KMEs is not known or not clear to those that maintain them, the CME's mitigation capabilities will erode.

Since there may be multiple KMEs that maintain the integrity of each CME, a facility should consider developing common groupings of KMEs that support CMEs. For example, a gas plant might document all the KMEs that support mechanical integrity across all the plant's critical equipment which would include the plant's critical instrument calibration and test frequency, pressure vessel integrity program, piping integrity program, etc.

Each KME could be mapped to a plant procedure or other documents, and the group number could be referenced on the PHR and Risk Register (attached complete group list to the PHR and Risk Register) rather than listing them every time the documents require them. The unique KMEs should be listed in addition to the group(s).

The supporting documents that maintain KMEs should be flagged so that those revising the documents are alerted to their significance (i.e., the MOC team would know to include this in their analysis).

3.2 CONTROL SYSTEM CMES

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3.2.1 Application of CMEs to Control System

Control Systems, and CMEs in the context of this Guideline, consist of:

- Basic Process Control Systems (BPCS),
- Safety Instrumented Systems (SIS),
- Safety Instrumented Functions (SIF),
- Instrumented Protective Systems (IPS),

as described below in this section.

Appendix 2 describes the general treatment of Control Systems and assigning CMEs in Process Hazard Reviews (PHRs).

3.2.1.1 BPCS

BPCS are commonly DCS or PLC-based Control Systems but may also be systems consisting of switches, relays, wiring and final control elements. BPCS primarily serve to perform control functions but may also provide safety protective functions. Historically, BPCS have not been designed or maintained to be as reliable as a dedicated emergency shutdown system and, therefore, only one (1) loop within a BPCS can usually be given credit as a CME provided it meets the criteria of Section 3.2.2. Also, a single BPCS loop can only be counted as one (1) CME, unlike an SIS loop. For modern BPCS utilizing redundant component designs and robust management practices, two (2) CMEs may be credited for two separate BPCS loops subject to the requirements in Section 3.2.3 below.

3.2.1.2 Safety Instrumented Systems (SIS)

SIS are highly reliable and should only rarely experience a dangerous (undetected) failure (fails to function when required with no indication of the impending failure) or safe failure (spurious trip). The functions within an SIS are referred to as Safety Instrumented Functions (SIF).

3.2.1.3 Safety Instrumented Functions (SIF)

As denoted by CME PR-1, CMEs for SIF are equated to the specified Safety Integrity Level (SIL). A higher SIL equates to a lower Probability of Failure on Demand (PFD) of a dangerous failure. Therefore, a SIL 1 function would be equivalent to 1 CME; SIL 2 would equal 2 CMEs, and so on. By design, a single SIF loop (e.g., high pressure shutdown) may be credited 2 CMEs if the SIF is designed, operated, and maintained to SIL 2 or higher. In contrast, a single BPCS loop can never be credited with more than 1 CME. An Event Scenario in which the initiating event involves failure of a SIF, such a spurious trip in which an ESD valve suddenly closes during normal process operations, is considered a very low probability event and, therefore, CME IF-6 is available to be credited for this situation.

The CMEs to be credited for SIF are predicated on these systems being appropriately operated and maintained in accordance with the requirements of ANSI/ISA 84.00.01 / IEC 61511 Mod, Safety Instrumented Systems for the Process Industry Sector. Effective Implementation of SIS requirements necessitates a system to manage the design, operation, maintenance, and assessment of these systems to ensure that they achieve the specified safety integrity level.

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Further, all associated documentation needs to be maintained such as the Safety Requirements Specification, SIS design, engineering, installation, testing, plan for SIS operation, maintenance, modifications, verification, and functional safety assessment.

3.2.1.4 IPS

As the terms “SIS” and “SIF” imply adherence to ANSI/ISA 84.00.01 / IEC 61511 Mod, those systems which may have been designed to these industry standards but have not been operated or specifically maintained as such are given the term “Instrumented Protective System” (IPS) to differentiate between the two. In effect, IPSs are SIS for which the specified reliability cannot be demonstrated to the extent required by the ISA/IEC standards. Therefore, these systems must be treated similarly to BPCS in terms of crediting CMEs. In order to credit two (2) IPS shutdown loops as CMEs for an Event Scenario, the criteria of Section 3.2.3 and Functional Assessment criteria of Section 3.2.4 below should be evaluated and met. Similar to a BPCS, multiple CMEs cannot be credited to a single IPS loop.

If an initiating event (Event Scenario cause) involves a BPCS loop failure, then up to two (2) IPS loops may still be credited as CMEs provided these are totally separate from the BPCS, refer to Appendix 3 below for the basis of CME control of system loops. A different BPCS loop may also be credited as a CME subject to the criteria of Sections 3.2.2

If an initiating event involves an IPS loop failure, then only one (1) independent IPS loop may be credited as a CME and no credit can be given for low probability of an IPS failure initiating the Event Scenario

3.2.2 Criteria for Crediting One (1) BPCS Loop as a CME In An Event Scenario

The BPCS loop to be credited as a CME (PR-6) must be effective in preventing or mitigating the Event Scenario as qualitatively judged by the PHR team. Also, the BPCS loop must be reliable as evident by Key Mitigation Elements (KMEs) in place such as periodic maintenance, inspection, and testing. Further, there should not be a history of loop failures nor should there be downgraded conditions or situations that would reduce the effectiveness or reliability of the loop.

If failure of a BPCS loop is the initiating event (cause) for an Event Scenario, then for a different BPCS loop to be credited as a CME the following criteria apply:

1. The BPCS loop must be functionally independent of the loop considered to have failed in terms of utilizing a separate sensor and final control element.
2. There must be no common cause failure mechanisms involving the sensors or final control elements that would prevent the independent BPCS loop from performing its intended function (e.g., a common cause failure might be a separate sensor that is mounted in a vessel where it also may be subject to being plugged and fail to function).
3. There must be no history of a high failure rate for loop components, including I/O cards, logic solvers, and wiring that may be common to both the failed BPCS loop (initiating event) and the BPCS loop claimed as a CME.
4. The independent BPCS loop must be designed such that the fail position of the final control element (valve, switch, etc.), is the same as the desired action to prevent or mitigate the Event Scenario.

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5. The independent BPCS loop credited as a CME must be verified (proven by testing) for “de-energize to trip” mode upon loss of utilities (i.e., electric, air, or other motive force), or open circuit failures. Documentation must exist and be available to PHR teams to support this claim.
6. The independent BPCS loop credited as a CME must be verified (proven by testing) for “loss-of-signal to trip” mode upon loss of communications between field devices and logic solver, or open circuit failures. Documentation must exist and be available to PHR teams to support this claim.
7. If the “independent” BPCS loop receives a compensation signal from the failed BPCS loop (e.g., temperature compensation), then the independent loop should not be credited as a CME unless the PHR team has determined that the compensation effects are inconsequential to its function as a CME.

3.2.3 Operation & Maintenance Criteria for Crediting Two (2) BPCS Loops as CMEs In An Event Scenario

Criteria of this section are intended to be assessed by PHR teams with input from an Automation Specialist in order to credit two (2) BPCS loops as CMEs within a common BPCS. Additionally, the criteria of Section 3.2.4 – Functional Assessment should be met to credit two (2) BPCS control loops within a BPCS. Refer to Figures 1 – 4 for illustration of various configurations of control loops.

3.2.3.1 General Criteria

If the initiating event for the scenario involves a BPCS loop, then no more than one (1) CME may be credited for an independent loop in the same BPCS. Refer to Section 3.2.2 above for criteria for crediting a single CME for a BPCS loop.

1. If the initiating event for the scenario involves a BPCS loop, then no more than one (1) CME may be credited for an independent loop in the same BPCS. Refer to Section 3.2.2 for criteria for crediting a single CME for a BPCS loop.
2. The control loops within the BPCS credited as CMEs for an Event Scenario must be independent of each other with respect to sensors and final control elements.
3. The logic solver may be common to the control loops but the criteria of Section 3.2.4.2 below should be met.
4. Motor starters and breakers may be common to BPCS loops credited as CMEs for the same Event Scenario.
5. More than one CME cannot be credited to loops in which the input Voting Logic is a 2oo2 (2 out of 2) configuration or 3oo3 (see Section 3.2.6).
6. There must be no common cause failure mechanisms involving the sensors or final control elements that would prevent the independent BPCS loops from performing their intended function (e.g., a common cause failure might be a separate sensor that is mounted in a vessel where it also may be subject to being plugged and fail to function).
7. Each of the BPCS loops credited as CMEs (PR-6) for the same Event Scenario must be sufficiently effective in preventing or mitigating the Event Scenario as qualitatively judged by the PHR team. Also, the BPCS loops

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must be sufficiently reliable as evident by Key Mitigation Elements (KMEs) in place such as periodic maintenance, inspection, and testing. Further, there should not be a history of loop failures nor should there be downgraded conditions or situations that would reduce the effectiveness or reliability of the loops.

8. Each of the BPCS loops credited as CMEs for the same Event Scenario must be verified to fail to a safe state with respect to all system components. (i.e. designed such that the fail position of the final control element (valve, circuit breaker, etc.) is the same as the desired action to prevent or mitigate the Event Scenario). Documentation must exist and be available to PHR teams to support this claim.
9. Each of the BPCS loops credited as CMEs for the same Event Scenario must be verified (proven by testing) for “de-energize to trip” mode upon loss of utilities (i.e., electric, air, or other motive force), or open circuit failures. Documentation must exist and be available to PHR teams to support this claim.
10. Each of the BPCS loops credited as CMEs for the same Event Scenario must be verified (proven by testing) for “loss-of-signal to trip” mode upon loss of communications between field devices and logic solver, or open circuit failures. Documentation must exist and be available to PHR teams to support this claim.
11. If an “independent” BPCS loop is compensated by a function of another BPCS loop credited as a CME (e.g., temperature compensation), then the independent loop should not be credited as a CME unless the PHR team has determined that the compensation effects are inconsequential to its function as a CME.

3.2.3.2 Input / Output Card Criteria

1. The input / output card for each BPCS loop credited as CMEs for the same Event Scenario must be independent from one another unless the installation includes one-for-one redundant input cards consisting of a primary card with a “hot swappable” secondary card.
2. Transfer to the secondary input / output card must be automatic, “bumpless”, and cannot be defeated (unless when required for maintenance or repair activities). After primary failure, the secondary input card should be fully operational within one (1) second of the primary failure.
3. A failure in the primary or secondary input / output card should be promptly and automatically alarmed.
4. An input / output card for any BPCS loop credited as a CME may not be Remote I.O. (Guideline for use of remote I.O. cards to be published by the Idd El Shargi Automation Technology group.)

3.2.4 Functional Assessment Criteria

The criteria of this section pertain to the Control System architecture and are intended to be assessed by Automation Specialists. All the Functional Assessment criteria should be met to credit two (2) control loops as CMEs within a BPCS as follows:

3.2.4.1 General Criteria

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1. The Control System has been maintained to be current in terms of firmware software (critical patches) upgrades and does not have a history of failures (power, I/O cards, etc.).
2. UPS backup power is provided for the Control System and is kept in a reliable state through periodic maintenance, inspection, and testing.

3.2.4.2 Logic Solver Criteria

1. Control systems should have one-for-one redundant logic solvers consisting of a primary with a “hot swappable” secondary.
2. Transfer to the secondary logic solver should be automatic, “bumpless”, and cannot be defeated (unless when required for maintenance or repair activities). After primary failure, the secondary logic solver should be fully operational within one (1) second of the primary failure.
3. The secondary logic solver should contain all the information of the primary within one (1) second and be capable of continuing the process exactly as it was when it was switched in without interruption or process upset. All control functions including analog control, logical functions, programs, set points, and tuning parameters should continue as before and any changes entered through a console should be current in the secondary unit. The secondary logic solver should be synchronized with the primary and be able to continue the sequence programs exactly where the primary logic solver stopped.
4. After a long power outage, all outputs should assume a programmed power recovery state (i.e., “safe state”), upon power resumption.
5. A failure in the primary or secondary logic solver should be promptly and automatically alarmed.
6. No single failure can affect the operation of both the primary and the secondary logic solver. A failure can be a hardware failure, software failure, power supply failure, or the loss of one AC feeder in dual feeder systems.
7. Logic solvers should be unaffected by failure of other devices connected to the communications system including other logic solvers.
8. The above criteria may be waived if another comparable approach is approved by Idd El Shargi Technical HSE Engineering.

3.2.4.3 Access & Security Criteria

1. Changes to Control System configuration and programmed logic, and adjustments or changes to alarm settings and controller algorithms, should be protected by password or keyed locks, and should only be made by designated individuals.
2. Where a personal computer is used to configure or program Control System logic, care must be taken to ensure that viruses do not exist that could corrupt servers on the DCS related networks. Configuration efforts should be made in full compliance with IT Policy, concerning digital information security.
3. Connections from the DCS network to external networks, such as a plant LAN, should be made in full compliance with Idd El Shargi IT Policy.
4. Changes to programmed logic and configuration of all points and alarms should be made in accordance with Idd El Shargi MOC Procedure QP-ISND-RM-304.

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5. The Business Unit has designated individuals who are authorized to make the changes identified above and the level of training required.

3.2.4.4 Communications

For DCS, sufficient equipment should exist to provide a fully redundant DCS network. The fail-over from the primary to secondary highway should normally be automatic and be alarmed; however, manual switchover capability should also exist. Means should exist to check the condition of either of the communication highways at all times. A single failure of either of the highways should not result in any system degradation.

3.2.5 Control Loop Figures

Figures 1 – 4 illustrate common control loop configurations

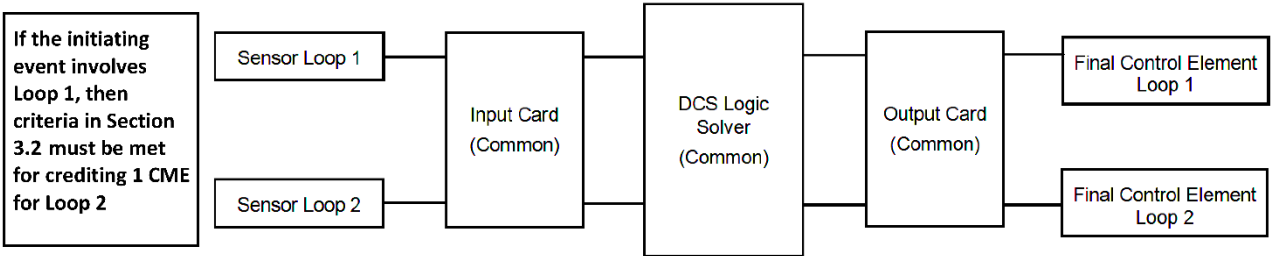


FIGURE 1. COMMON I/O CARDS AND LOGIC SOLVER FOR THE SAME SCENARIO (1CME)

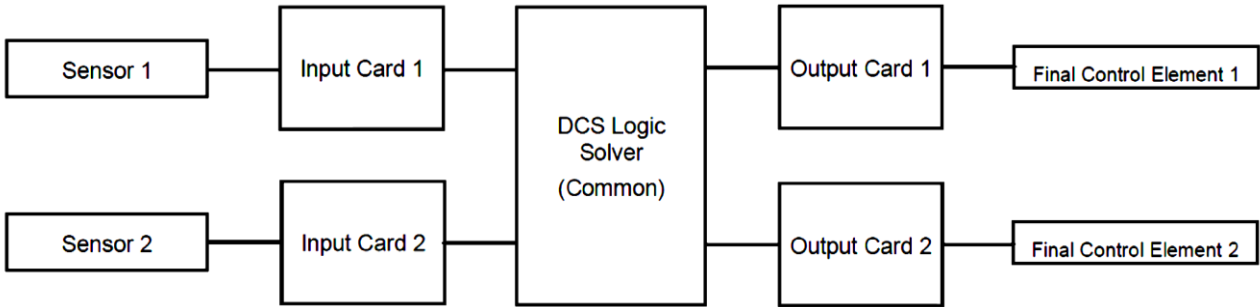


FIGURE 2. TWO INDEPENDENT LOOPS AND COMMON LOGIC SOLVER (2 CMES IF CRITERIA IN SECTIONS 3.2.3 AND 3.2.4 ARE MET)

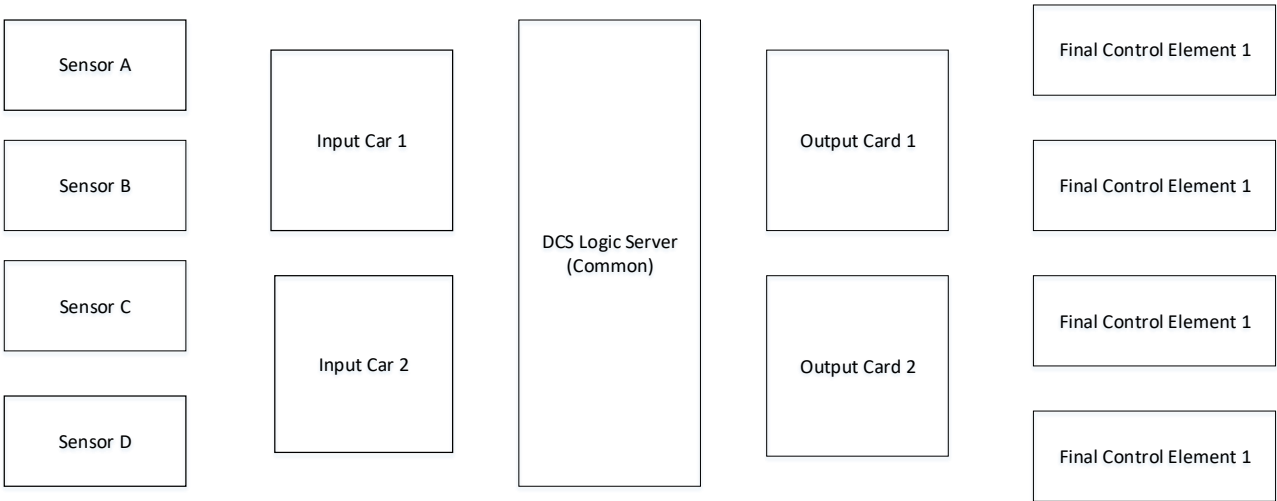


FIGURE 3. INDEPENDENT I/O, MULTIPLE SENSORS & FC ELEMENTS (CME IF CRITERIA IN SECTIONS 3.2.3 AND 3.2.4 ARE MET)

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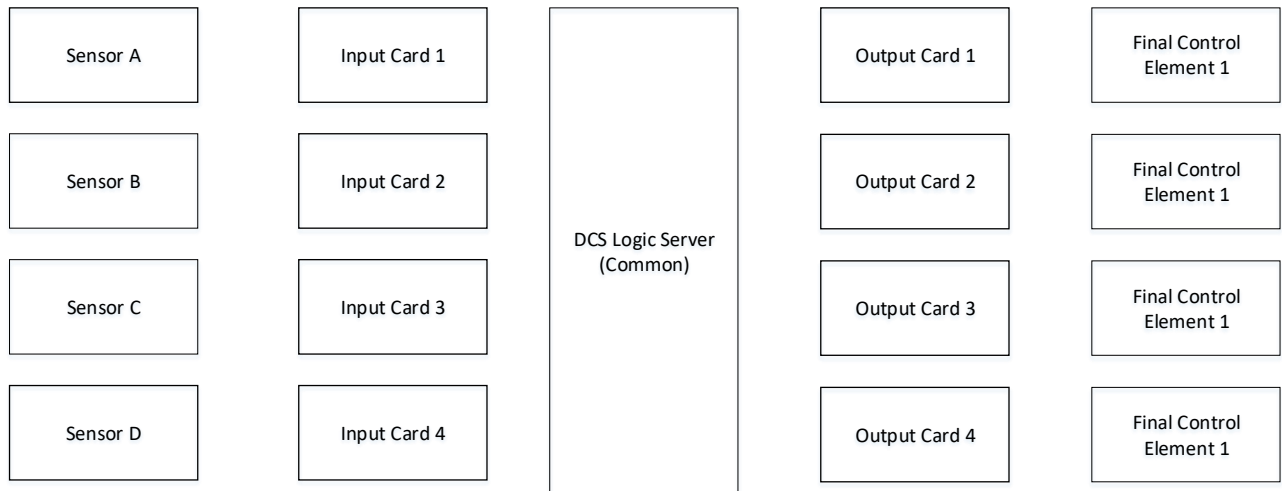


FIGURE 4. FOUR INDEPENDENT LOOPS WITH COMMON LOGIC SOLVER

(2 CMES IF CRITERIA IN SECTIONS 3.2.3 AND 3.2.4 ARE MET)

3.2.6 Considerations for New, Replacement & Upgraded BPCS Criteria

The design of Control Systems should strike a balance between safe operations (prevention of dangerous system or component failures), overall reliability/production and costs. Prevention of system failures through independence of components should take precedence over attaining a high level of redundancy.

In general, safe operations should be addressed through inherently safer design practices and non-control system prevention/mitigating CMEs. When these options are not available or do not provide the desired risk reduction, incorporation of independent BPCS design features that are supported by appropriate maintenance and management control as described in the previous sections may provide sufficient reliability to credit up to two (2) CMES. If further risk reduction is needed, then implementation of an independent Safety Instrumented System (SIS) may be warranted. However, the life cycle cost of an SIS should be evaluated against other potential mitigation options before choosing SIS.

This section discusses control device configurations that highlight the importance of building independence while at the same time increasing reliability. The basic assumption taken in all configurations is that the reliability of the logic solver is much higher than the reliability of the input and output components and that the Mean Time to Repair (MTTR) is much less than the test interval (TI) for the system components. There are several options for a Control System configuration as shown below. Standard nomenclature of MooN (e.g., 1 out of 2) is used in these diagrams to illustrate the Voting Logic for input and the final element configuration for outputs.

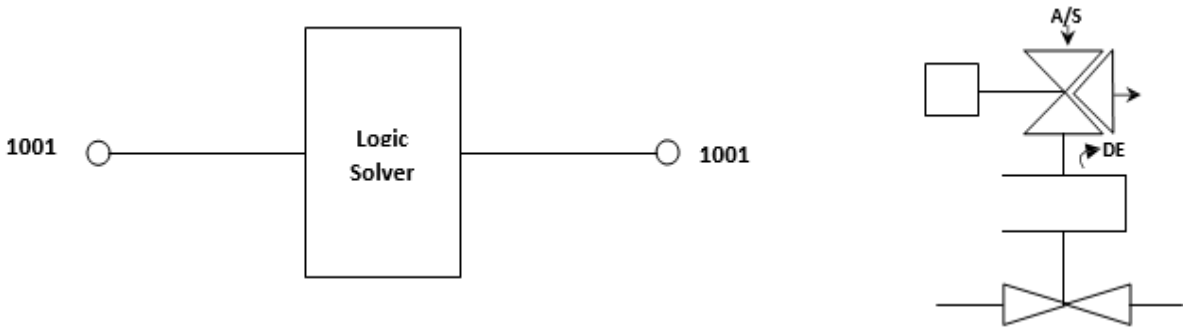


FIGURE 5. 1001 INPUT AND 1001 OUTPUT

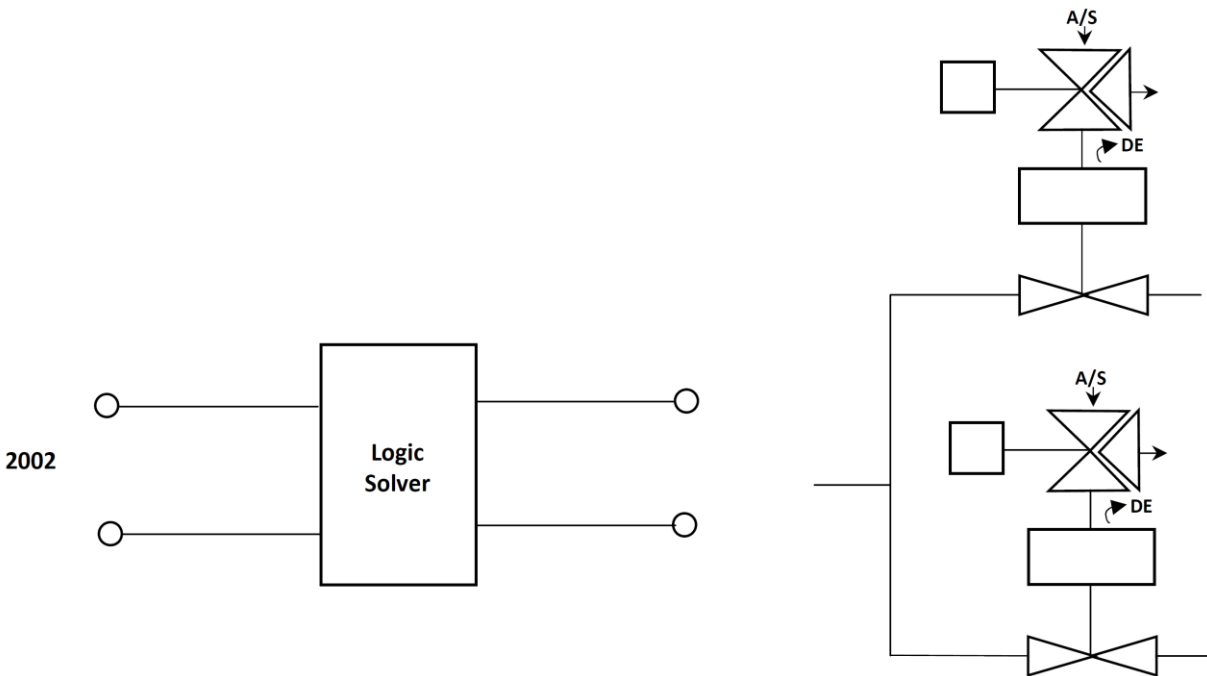


FIGURE 6. 2002 INPUT AND 2002 OUTPUT

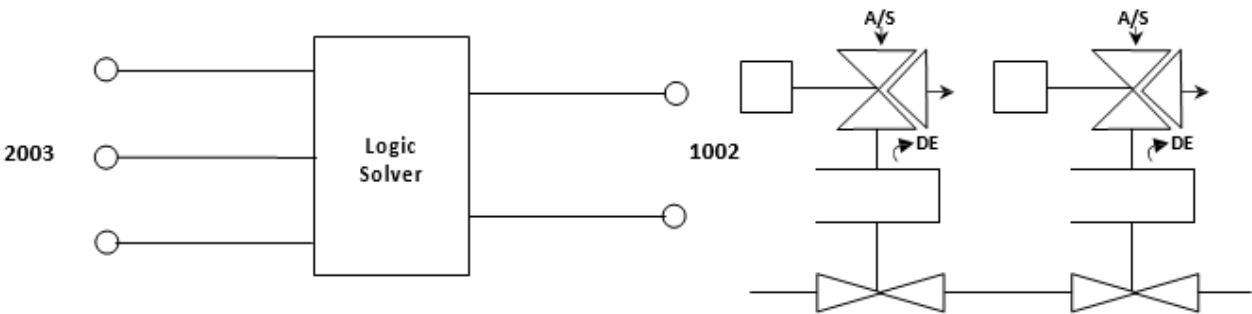


FIGURE 7. 2003 INPUT AND 1002 OUTPUT

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One can compare these control function configurations using fictitious numbers to derive the Probability of Failure on Demand (PFD) for dangerous failures and the Mean Time to Failure (MTTF) for spurious trips (safe failures). Table 1 defines the variables and values used for comparison calculations and Table 2 shows the resulting PFD and MTTF values for five (5) common control system configurations using simplified equations. Very general and simple assumptions used here would not typically be utilized in a detailed reliability study or SIL determination.

TABLE 1. VALUE AND VARIABLES USED FOR SYSTEM CONFIGURATION EXAMPLE

Symbol or Abbreviation	Description	Units	Value
λ_d	Dangerous (Inhibiting) Failure Rate	Per year	0.1
λ_s	Safe (Initiating) Failure Rate	Per year	0.2
TI	Test Interval	Years	1
MTTR	Mean Time to Repair	Years	0.00274
STR	Spurious Trip Rate	Per year	Calculated
MTTF	Mean Time To Failure, safe	Years	Calculated
PFD	Probability of Failure on Demand	Unitless	Calculated

TABLE 2. EXAMPLE PFD & MTTF FOR VARIOUS CONTROL CONFIGURATIONS

Configuration	PFD Equation	PFD	STR Equation	STR	MTTF (1 / STR)
1001	$\lambda_d \times TI/2$	0.05	λ_s	0.2	5.00
1002	$[(\lambda_d)^2 \times (TI)^2] / 3$	0.003	$2 \times \lambda_s$	0.4	2.50
2002	$\lambda_d \times TI$	0.01	$2 \times (\lambda_s)^2 \times MTTR$	0.00022	4,545.00
1003	$[(\lambda_d)^3 \times (TI)^3] / 4$	0.00025	$3 \times \lambda_s$	0.6	1.67
2003	$(\lambda_d)^2 \times (TI)^2$	0.01	$6 \times (\lambda_s)^2 \times MTTR$	0.00066	1,520.00

1001 Voting Configuration

A common process control loop is 1001 configuration, providing one input sensor, logic solver (electronic or human), and one output such as a control valve. The PFD in this example is 0.05. CMEs are expected to reduce the likelihood of an Event Scenario by 10-1 to 10-2 (0.1 to 0.01). Therefore, in this example in which the control system has a PFD of

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0.05, the process control loop may be credited as 1 CME if properly supported by KMEs (e.g., testing) to ensure that the system is reliable. If the test interval were extended to more than 2 years in this system, the PFD would be greater than 0.1 and should not be credited as a CME.

Note: This is not to imply that test intervals should not exceed 2 years, but rather is shown here to emphasize the importance of testing to ensure reliability.

1002 Voting Configuration

This configuration provides a significant reduction in the PFD but at the risk of more spurious trips. A process control function of this configuration which has independent input sensors, final elements, input/output cards and logic solvers, (or redundant input/output cards and logic solvers) utilizing this Voting Logic could be credited as two CMEs provided that the criteria in Sections 3.2.3 – 3.2.5 are met.

2002 Voting Configuration

When on-line reliability is important for operations, this configuration is desirable because the spurious trip rate is very low and corresponding MTTFs is very great. However, it has a higher PFD than the 1001 configuration because it requires both devices to detect a process deviation before taking action such as initiating a process shutdown. Therefore, this configuration should not be assigned more than 1 CME.

1003 Voting Configuration

This configuration provides a high degree of safety protection as the PFD is very low in this example, equivalent to SIL 3 for Safety Instrumented Systems (SIS). However, the safe failure rate is the highest of these configurations, meaning it will have the highest probability of a spurious trip. Despite the low PFD shown in this example, the actual failure rate may be significantly higher due to systematic failures introduced by errors relating to specifications, design, and implementation in both hardware and software. A process control function of this configuration which has independent input sensors, final elements, input/output cards and logic solvers (or redundant input/output cards and logic solvers) utilizing this Voting Logic could be credited as two CMEs provided that the criteria in Sections 3.2.3 – 3.2.5 are met.

2003 Voting Configuration

The Voting Logic of 2003 provides good safety protection by substantially reducing the PFD over a 1001 configuration while significantly reducing the possibility of a spurious trip. A process control function of this configuration which has independent input sensors, final elements, input/output cards and logic solvers (or redundant input/output cards and logic solvers), utilizing this Voting Logic could be credited as two CMEs provided that the criteria in Sections 3.2.3 – 3.2.5 are met.

Maximum Number of CMEs Allowed for BPCS Configurations.

The maximum number of CMEs credits allowed for various BPCS configurations is summarized in the following table.

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TABLE 3. MAXIMUM NUMBER OF CME CREDITS ALLOWED

Configuration	Maximum # of CMEs
1001	1
1002	2
2002	1
1003	2
2003	2

Note: BPCS must meet the criteria specified in Sections 3.2.3 – 3.2.5 to be credited more than one CME.

3.2.7 Testing Requirement – Specific & Time-Based Maintenance

Function testing shall be performed according to Emergency Shutdown Systems Maintenance, Testing & Inspection Procedure, QP-ISND-PRC-PRD-621.

4.0 REFERENCES TO OTHER DOCUMENTS

TABLE 4: REFERENCES TO OTHER DOCUMENTS

Document Title	Document Number	Internal/External Document (if any)	Document Referencing
Risk Evaluation and Response Procedure	IS-HSE-PRC-301	Internal	Upwards
Process Hazard Review Procedure	IS-HSE-PRC-308	Internal	Upwards
Risk Management Mechanical Integrity Procedure	IS-HSE-PRC-309	Internal	Sideways
Hazard & Operability (HAZOP) Review Guideline	IS-HSE-GDL-318	Internal	Sideways
Evacuation, Escape, & Rescue Analysis (EERA) Guideline	IS-HSE-GDL-320	Internal	Sideways
What If Analysis Guideline	IS-HSE-GDL-316	Internal	Sideways
Safety Integrity Level (SIL) Guideline	IS-HSE-GDL-319	Internal	Sideways

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5.0 DEFINITIONS

TABLE 5: KEY TERMS DEFINED

Term	Definition
HAZOP	The application of a systematic critical examination, to the process and engineering intentions, of a facility to assess the hazard and operability potential of mal-operation or malfunction of individual items of equipment and the consequential effects on the facility as a whole.
PROCESS HAZARD ANALYSIS	An organized, thorough and systematic approach to identify and evaluate hazards associated with chemical processes and operations to enable their control.
What-If-Analysis	What-If Study is a structured brainstorming method of determining what can go wrong and judging the likelihood and consequences. The answers to these questions form the basis for making judgments regarding the acceptability of those risks and determining a recommended course of action for those risks judged to be unacceptable.
MANAGEMENT OF CHANGE (MOC)	MOC is a formal process that systematically manages permanent, temporary or emergency changes to the organizations, operations, process equipment, technologies, procedures or chemicals that could impact the original design, operations and maintenance criteria, and/or environmental performance.
Critical Mitigation Element (CME)	An independent and reliable device or system that is capable of: • Preventing initiation of an Event Scenario, • Preventing loss of Hazardous Material containment, or • Substantially mitigating one of an Event Scenario's Consequences. Failure of one CME should not prevent another CME from performing its function. CME should be maintained to the reliability level assumed in the risk analysis.

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APPENDIX

APPENDIX A - CME GROUPS

Fire or Explosion Event Scenarios - Credit for low probability initiating events

(A CME credit can only be taken once – not in each CME Group, unless there is a totally independent system that accomplishes the mitigation)

CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
Relief Valve High Reliability (preventing an initiating event)	This CME group applies to Event Scenarios where the initiating event is a relief valve that fails in full open position. If in clean service (e.g., natural gas) and all applicable KMEs apply (e.g., testing/maintenance), a CME is appropriate.	1
Mechanical Integrity – Prevention and Detection	This CME credit is applicable where the initiating event is corrosion, erosion, or other material damage mechanism and a system is in place to prevent or significantly mitigate the damage mechanism. Examples include corrosion inhibitor systems, coatings/linings, and use of a material that has a significantly lower corrosion rate than normally used for the process. To take credit for this CME, all necessary KMEs must be in place to ensure that the addition of corrosion inhibitor is functioning as intended; coating/linings are appropriate designed, installed, inspected and maintained; and/or the material selected will demonstrate lower corrosion rates. The KMEs would include, but not be limited to, use of corrosion coupons to monitor corrosion rate, on- line sampling of corrosion inhibitor concentration, manual sampling and testing for corrosion inhibitor residual, chemical provider usage monitoring program, etc.	1
Mechanical Integrity – Increased Design	This CME credit will be applied for mechanical design that is significantly above basic requirements and where failure could only result in a slow degradation with ample time to detect degradation before failure. The actual wall thickness shall be at least 50% greater	1

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CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
Margin for Detection Before Failure	than calculated for mechanical design for the rated operating pressure and temperature or 50% greater than the structural required thickness (whichever is greater). To credit this CME, the minimum corrosion allowance shall be at least 0.125" or greater. All necessary KMEs shall be in place to ensure the integrity of the piping/vessel is maintained. The KMEs shall include, but not be limited to, periodic wall thickness measurements, corrosion probes, smart pigging or any other device to determine corrosion rates and remaining life. Valid KME's for underground piping are cathodic protection, periodic hydrostatic testing, smart pigging etc. Credit for this CME could also apply to the use of stronger, significantly more robust materials than the design specification called for, such as using stainless steel where the design called for carbon steel.	
Mechanical Integrity Design-Self - Announcing	This CME credit applies to prevention of loss of containment from an excavation striking a pipeline (incidental contact) underneath a concrete/asphalt road that is encased in slurry used in directional boring. The pipe's concrete slurry encasement would warn the person performing the excavation much like red dyed concrete would warn of a buried electrical conduit and dye/ribbon would warn of a buried pipeline.	1
Emergency Shutdown Devices (ESDs), Burner Management System, and other Safety Instrumented Systems (SIS)	For Event Scenarios where an ESD, BMS, or other SIS failure is the initiating event, this CME could be appropriate. This CME credit(s) is taken when there is a low probability of a control system initiating the event scenario (e.g., a control system spurious trip that causes an unsafe shut down sequence that initiates the Event Scenario). All necessary KMEs shall be in-place to ensure the systems will not degrade to a level where failure is likely to occur. See Section 3.2 of this Guideline concerning Control Systems CMEs should full SIS requirements not be met.	SIL 1 = 1 SIL 2 = 2 SIL 3 = 3 Ref. Section 3.1.3.1
Car-Sealed or Locked Valve	For Event Scenarios where inadvertent opening/closing of a manual valve is the initiating event, this CME would be appropriate. A system shall be in place (documented list of car seals/locks and periodic checks) to verify that car-seals or locks are in place and valves are	1

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CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
	in the correct position to be credited as a CME. Car-seals shall be of substantial design so as not to be easily breakable (e.g., cables or chains). Inadvertent operation of valves that are not readily accessible or are so large as to be difficult to operate would typically be an incredible Event Scenario.	
Contingent Element – 90% Reliability of Contingent Mitigation	For Event Scenarios where two or more conditions shall be present in a compounded fashion for an Event Scenario to be possible (typically an enabling condition), an initiating event CME will be appropriate if one of those conditions is impaired <10% of any year. For example, a pump is run in certain configuration for 5 days out of a year and this configuration has features that could lead to an overpressure event. Or, 90% of the time during any year, drainage is available to take a flammable liquids spill away from exposing vessels with a BLEVE potential – drainage is not available when rainwater is being contained for environmental mitigations 10% or less of the time. This is in contrast to multiple jeopardy events, where two or more unrelated failures shall occur simultaneously (i.e. not in a compounded fashion) and such events are generally deemed incredible due to very low probability of occurrence.	1
Redundancy	For Event Scenarios where 100% reliable redundancy is available to prevent the initiating event, this CME would be appropriate. For example, two completely redundant, 100% capacity, automatic start up instrument air compressors are available – if one fails, the backup redundant compressor would prevent the initiating event. For 200% totally independent redundancy, 2 CMEs are available. 300%, 3 CMEs, etc. Caution: All CME convention rules shall be met.	1 (2+) Ref. Section 3.1.3.1
Bull Plug Program	This applies to Event Scenarios such as “Vent or Drain Valve Left Open in Error”, or similarly a “Vent or Drain Valve Leaks By” where a bull plug would prevent the scenario from occurring. If the site has a program that assures that bull-plugs are installed and maintained	1

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CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
	at the end of open ended lines containing flammable or toxic materials, a CME would be appropriate. The KMEs for this CME would include a documented and audited program for routine monitoring, plug replacement, and LDAR detection (where applicable).	
Robust Process Isolation Barriers	This applies to Event Scenarios involving a potential inadvertent valve operation that could result in personnel exposure to high pressure or corrosive/toxic/flammable material, or downstream equipment damage due to high pressure or other undesired factors. For the CME to be credited, there must be a program in place that requires a documented valve status verification for each isolation by a secondary personnel. The KME for this CME includes a documented and audited isolation/permitting program, such as the Integrated Safe System of Work (ISSoW), incorporating clear instructions for secondary personnel checks. This CME cannot be credited with other CMEs involving Human Intervention. In addition, the CME cannot be credited if the inadvertent valve operation is postulated to take place after the system has been verified by the secondary personnel.	1

Fire or Explosion Event Scenarios - Prevent release or substantially reduce the Probability or Consequence of a release

CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
Emergency shutdown Devices (ESDs), Burner Management System, & other Safety Instrumented Systems (SIS)	This CME group consists of sensor(s), logic processor(s) and final element(s) that detect out-of- limit conditions and bring the process to a functionally safe state without human intervention. See Section 3.2 of this Guideline concerning Control Systems CME, should full SIS requirements not be met.	SIL 1 = 1 SIL 2 = 2 SIL 3 = 3 Ref. Section 3.1.3.1



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Innately Safe Design	This CME may be credited if it can be demonstrated that the design prevents this Event Scenario or Significantly mitigates the consequences. This CME cannot be credited in addition to other CMEs, such as the Innately Safer Mechanical Integrity Design CME (PR-7) below, unless the two are totally independent. Examples include: 1) a pump whose maximum discharge pressure does not exceed the design pressure of the downstream equipment, and 2) valve actuation time that is sufficiently slow to avoid an overpressure situation. Note: A short comment in the PHR shall be provided for future reviewers to understand the credit and the factors that could otherwise allow the Event Scenario to be considered “incredible”.	1
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CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
Human Intervention	A CME can be assigned when one or more system(s) or device(s) requires human intervention to take preventative or mitigative action. This CME can be assigned ONLY IF all of the following are true: 1. the system or device is monitored 100% of the time by personnel, 2. personnel have sufficient time to analyze the situation and take the required action that will prevent or Mitigate the Event Scenario, and 3. all necessary KME are in place, such as procedures, training, testing and drills. This CME can be credited in addition to an automatic control or ESD system CME if the Human Intervention system is totally independent¹ of these other systems. Offshore Specific A CME can be credited for offshore operations where human intervention in terms of escape could mitigate the consequences. This CME can be if assigned ONLY IF in addition to the above, all of the following are also true: 4. personnel working in the plant area have full knowledge of escape routes, 5. carry personal gas detectors, and have immediate access to BA sets.	1
Pressure Relief Devices (PRDs)	PRD's prevent systems from exceeding specified overpressures. For this CME to be applicable, potential consequences resulting from the PRD flow shall be evaluated (e.g., flammable gases released into congested areas). Effectiveness of PRDs is sensitive to service and experience. All necessary KMEs shall be in-place (e.g., testing, calibration, maintenance) to ensure the systems will not degrade to a level where failure is likely to occur.	1

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Scrubbers and Flares	This CME group constitutes an integral part of a pressure relieving system that provides a level of protection by Mitigating the Consequences associated with a flammable gas release. This CME can only be applied when credit for those systems is not taken as an integral element in another CME such as ESD.	1
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CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
Detection and Control Systems (Fire, LEL, Flow, Pressure, Temperature, Composition)	This CME group consists of sensor(s), logic processor(s) and final element(s) that detect out-of- limit conditions and bring the process to a functionally safe state. For CME credit, the system shall be totally independent of other CME credits. Typically, this is a Basic Process Control System (BPCS) such as a DCS, PLC, or relay logic control. See Section 3.2 of this Guideline on Idd El Shargi Control Systems CME for clarification.	1
Innately Safer Mechanical Integrity Design	Use of a material that is known to not be susceptible to corrosion, erosion, or other failure mechanisms by the process. Failure could only result from slow degradation with ample time to detect degradation before failure (i.e., multiple inspection cycles). To apply this CME all necessary KMEs shall be in-place to ensure the systems will not degrade to a level that failure could occur. Examples of KMEs to maintain the system design parameters include, as appropriate, cathodic protection, chemical corrosion program, a robust MI program for monitoring and testing at appropriate frequency, and MOC program.	1
Secondary Containment Systems (Dikes, Berms)	These systems include passive devices that are designed, maintained and demonstrated to provide a level of protection that substantially reduces potential effects on people, the environment and PD/LOR.	1

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Drainage – Directs Hydrocarbons to Safe Location	This CME is applicable if hydrocarbons are directed away from other equipment, occupied structures, etc. to the extent that impacts and potential effects on people, the environment and PD/LOR are substantially reduced.	1
Car-Sealed or Locked Valve	For Event Scenarios where prevention of an inadvertent opening/closing of a manual valve would prevent the scenario, this CME would be appropriate. A system shall be in place (documented list of car seals/locks and periodic checks) to verify that car-seals or locks are in place and valves are in the correct position to be credited as a CME. Car-seals shall be of substantial design so as not to be easily breakable (e.g., cables or chains). Inadvertent operation of valves that are not readily accessible or are so large as to be difficult to operate would typically be an incredible Event Scenario. This CME cannot be credited if the valve is the same valve for which credit is taken in IF 7.	1
Check Valve	This CME will be applied for prevention or substantial reduction in backflow if judged to be effective (e.g., type of service, MI program, etc.) in mitigating the event scenario. Check valves shall not be credited as a CME for preventing pressure due to the potential for leak-by.	1

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CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Name
Concrete Barriers / Bollards and Other Physical Barriers	This CME will be applied if the barriers are sufficiently robust to prevent the event scenario.	1
Open Vent or Overflow Line	This CME will be credited for passive devices that are designed to relieve material to a safe or contained location. Any intervening block valves shall be car-sealed or locked to ensure functionality.	1
Sampling (Manual)	This CME is applicable if sufficient time exists to analyze multiple samples and take corrective actions so as to be considered reliable and effective. This CME cannot be credited with other CMEs involving Human Intervention.	1
Spare Equipment (Manual Startup or Activation)	This CME is applicable if sufficient time exists to bring the equipment on stream to prevent or mitigate the Event Scenario. The Event Scenario shall not involve a common cause failure (e.g., loss of power would affect both an operational and spare pump). This CME cannot be credited with other CMEs involving Human Intervention.	1
Spare Equipment (Automatic Activation)	This CME is applicable if the Event Scenario does not involve a common cause failure (e.g., loss of power would affect both an operational and spare pump).	1

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Fire or Explosion Event Scenarios - Prevention of Ignition or Explosion

CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
Elimination of Ignition Sources	This CME is applicable if all fixed sources of ignition have been eliminated through Electrical Classification, Bonding and Grounding, separation from fired equipment, separation from engine air intakes, etc. OR, wind rose demonstrates that wind direction will prevent ignition 90% of the time during a year .In addition, KMEs shall be in place to support fixed systems (e.g., maintenance and testing) and transient ignition sources (e.g., permit to work, hot work and MOC programs).	1
Explosion Protective Spacing/Layout	If wind rose (i.e. a plot of historic wind speed and direction at a particular location) demonstrates that the Event Scenario vapor release will not drift into a congested area 90% of a year, a CME can be credited for explosion consequence.	1
Tank Gas Blanketing	This CME is a design measure to prevent an explosion by elimination oxygen within the vapor space of tanks containing hydrocarbons. The blanket must be an engineered system using inert or other gases (typically methane) that will keep the vapor space outside the flammable range under all tank operating conditions. Appropriate KMEs must be applied to insure that the blanketing system is properly designed, operated and maintained as HSE Critical Equipment.	1

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Fire or Explosion Event Scenarios - Mitigate People Consequences

CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
Layout/Spacing /Separation Distance	A CME can be credited if layout/spacing/separation distance either prevents significant injuries or allows people enough time to escape without significant injuries. An explanation shall be documented if this CME is applied.	1
Fire and Blast Walls or Hardened Buildings	This CME group consists of passive devices that are designed and maintained to provide a level of protection by Mitigating the Consequence. This CME can be credited if the system protects people from the fire or blast effects of an ignited release, or if the system provides time for people to evacuate to a location that protects them from significant injuries.	1
Fireproofing	This CME group consists of passive materials that are designed and maintained to reduce the rate of heat input to vessels, piping and structures in order to delay or prevent Consequences (e.g., to provide additional time for depressurizing a vessel or firefighting). This CME can be credited if enough delay is provided to allow people to evacuate to a location that protects them from significant injuries.	1
Fire Protection	An active fire protection system can reduce the rate of heat input to vessels, piping and structures and potentially delay or prevent Consequences (e.g., to provide additional time for depressurizing a vessel or firefighting) or can protect people from significant injuries during their escape. This CME can be credited if enough delay or exit protection is provided to allow people to evacuate to a location that protects them from significant injuries.	1
Alert and Escape System/Factors	If people potentially impacted by the Event Scenario are immediately alerted of the release and are located in a building or otherwise protected from immediate consequences and they have a protected escape route (e.g., building or structure to mitigate thermal radiation exposure), a CME credit can be taken for applicable Event Scenario Consequences.	1

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CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
	This CME was removed as it was an error in earlier version.	
Contingent Element – Receptor Is Not Present > 90% of the Time	This CME will be credited for potential injury to personnel if personnel rarely (< 10% of the time) would be in the area affected by the event scenario. Example include a process controls initiated excursion at a flare or a tank farm. If personnel are in such portions of a facility on a limited basis, then a CME credit can be applied to account for lower probability of personnel impact.	1

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Fire or Explosion Event Scenarios - Mitigate Environmental Consequences

CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
Secondary Containment Systems (Dikes, Berms)	This CME group consists of passive devices that are designed and maintained to provide a level of protection by Mitigating the Consequence.	1
Contingent Element – Receptor Is Not Present > 90% of the Time	This CME will be credited if data can demonstrate the condition that could result in the Event Scenario's Consequences exists less than 10% of the time (i.e., the event would have to occur at the same time that another condition was in effect but that condition is only applicable less than 10% of the time). Example: A tank overflows during the limited window of time the tank dike's storm drain is open (which is open less than 10% of the time) allowing a spill to reach a water receptor.	1
Human Intervention – Environmental	This CME is applicable for ERP / SPCC Plan with human intervention to prevent or mitigate an Event Scenario. A CME can also be credited for emergency response / spill plans if the spill will be detected, equipment and manpower are always present (>90% of the time) and the scenario will develop slowly enough for the spill/release to be intercepted before it reaches the receptor (e.g., scenario is that oil could be spilled at a rate that will take an hour to reach a stream but there is a very high confidence that the event would be detected and booms would be laid in less than an hour to prevent any significant amount of oil from reaching the stream). This CME cannot be used in conjunction with any other CME that requires human intervention.	1

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Fire or Explosion Event Scenarios - Mitigate PD/LOR

CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
Layout/Spacing/Separation Distance	This CME group refers to passive means of separating a Hazard from potential Consequence areas which, as a result, Mitigates the Consequence. This CME will be credited if it can be demonstrated (typically through a consequence analysis) that the layout/spacing/separation distance substantially reduces PD/LOR. A consequence analysis shall be used to evaluate minimum spacing vs. optimal loss prevention separation; intensity (LP vs. HP gas); duration (volume/storage); and impact of a fire event (e.g. electrical & instrumentation vs. pipe only rack, or pipe rack configuration, ground sleeper, single or multi-level racks, and simple 'flash drum' vs. contactor absorber).	1
Fire and Blast Walls	This CME group consists of passive devices that are designed and maintained to provide a level of protection by Mitigating the Consequence. CME credit is applicable if it can be demonstrated and documented (typically a consequence analysis) that PD/LOR will be substantially reduced as a result of the devices. A consequence analysis shall be used to evaluate well rating (hours/overpressure) against the postulated event. Fire wall shall be "H" or UL High Rise rated for duration (0.5-4 hr). Blast walls shall be supported by blast modelling.	1
Fireproofing	This CME groups refers to passive materials that are designed and maintained to reduce the rate of heat input to vessels, piping and structures in order to delay or prevent Consequences (e.g., to provide additional time for depressurizing a vessel or firefighting). A CME credit can be taken if it can be demonstrated and documented that PD/LOR will be substantially reduced as a result of these materials. Demonstration shall include a Fire Proofing Assessment based on API 2218, and shall consider fire duration/intensity, escalation (potential collapse) and recovery/replacement time of plant/equipment.	1

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CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
Fire Protection	An active fire protection system can reduce the rate of heat input to vessels, piping and structures and potentially delay or prevent Consequences (e.g., to provide additional time for depressurizing a vessel or firefighting). A CME credit can be taken if it can be demonstrated and documented that PD/LOR will be substantially reduced as a result. The demonstration Assessment shall include type, nature and effectiveness of active systems (foam vs. water only), automatic vs. manual, ability to activate (sufficient trained operators to manually operate systems), adequacy of application rates, duration of supply, reliability and standards compliance (e.g. NFPA 20).	1
Process Isolation Valves	If the hydrocarbon inventory holdup within the process equipment and piping can be automatically isolated (or if manual, reliably and quickly) to substantially reduce the amount of hydrocarbons available to fuel an explosion or fire, a CME credit will be taken. This CME is only applicable if it is independent of shutdown system, process controls, or human interventions that have already been claimed as CMEs.	1

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Toxic Vapor OR Gas Release Event Scenarios - Credit for low probability initiating events

(A CME credit can only be taken once – not in each CME Group, unless there is a totally independent system that accomplishes the mitigation)

CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
Relief Valve High Reliability (preventing an initiating event)	This CME group applies to Event Scenarios where the initiating event is a relief valve that fails in full open position. If in clean service (e.g., natural gas) and all applicable KMEs apply (e.g., testing/maintenance per Idd El Shargi MI procedure IS-HSE-PRC-309), a CME is appropriate.	1
Deleted		
Mechanical Integrity – Prevention and Detection	This CME credit is applicable where the initiating event is corrosion, erosion, or other material damage mechanism and a system is in place to prevent or significantly mitigate the damage mechanism. Examples include corrosion inhibitor systems, coatings/linings, and use of a material that has a significantly lower corrosion rate than normally used for the process. To take credit for this CME, all necessary KMEs shall be in place to ensure that the addition of corrosion inhibitor is functioning as intended; coating/linings are appropriately designed, installed, inspected and maintained; and/or the material selected will demonstrate lower corrosion rates. The KMEs would include, but not be limited to, use of corrosion coupons to monitor corrosion rate, on-line sampling of corrosion inhibitor concentration, manual sampling and testing for corrosion inhibitor residual, chemical provider usage monitoring program, etc.	1
Mechanical Integrity – Increased Design Margin For	This CME credit will be applied for mechanical design that is significantly above basic requirements and where failure could only result in a slow degradation with ample time to detect degradation before failure. The actual wall thickness shall be at least 50% greater than	1

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CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
Detection Before Failure	calculated for mechanical design for the rated operating pressure and temperature or 50% greater than the structural required thickness (whichever is greater). To credit this CME, the minimum corrosion allowance shall be at least 0.125" or greater. All necessary KMEs shall be in place to ensure the integrity of the piping/vessel is maintained. The KMEs shall include, but not be limited to, periodic wall thickness measurements, corrosion probes, smart pigging or any other device to determine corrosion rates and remaining life. Valid KME's for underground piping are cathodic protection, periodic hydrostatic testing, smart pigging etc. Credit for this CME could also apply to the use of stronger, significantly more robust materials than the design specification called for, such as using stainless steel where the design called for carbon steel.	
Mechanical Integrity Design – Self-Announcing	This CME credit applies to prevention of loss of containment from an excavation striking a pipeline (incidental contact) underneath a concrete/asphalt road that is encased in slurry used in directional boring. The pipe's concrete slurry encasement would warn the person performing the excavation much like red dyed concrete would warn of a buried electrical conduit and dye/ribbon would warn of a buried pipeline.	1
Emergency Shutdown Devices (ESDs), Burner Management System, and other Safety Instrumented Systems (SIS)	For Event Scenarios where an ESD, BMS, or other SIS failure is the initiating event, this CME could be appropriate. This CME credit(s) is taken when there is a low probability of a control system initiating the event scenario (e.g., a control system spurious trip that causes an unsafe shut down sequence that initiates the Event Scenario). All necessary KMEs shall be in-place to ensure the systems will not degrade to a level where failure is likely to occur. See Section 3.2 of this Guideline concerning Idd El Shargi Control Systems CMEs should full SIS requirements not be met.	SIL 1 = 1 SIL 2 = 2 SIL 3 = 3 Ref. Section 3.1.3.1

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CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
Car-Sealed or Locked Valve	For Event Scenarios where inadvertent opening/closing of a manual valve is the initiating event, this CME would be appropriate. A system shall be in place (documented list of car seals/locks and periodic checks) to verify that car-seals or locks are in place and valves are in the correct position to be credited as a CME. Car-seals shall be of substantial design so as not to be easily breakable (e.g., cables or chains). Inadvertent operation of valves that are not readily accessible or are so large as to be difficult to operate would typically be an incredible Event Scenario.	1
Contingent Element – 90% Reliability of Contingent Mitigation	For Event Scenarios where two or more conditions shall be present in a compounded fashion for an Event Scenario to be possible (typically an enabling condition), an initiating event CME will be appropriate if one of those conditions is impaired <10% of any year. For example, a pump is run in certain configuration for 5 days out of a year and this configuration has features that could lead to an overpressure event. Or, 90% of the time during any year, drainage is available to take a flammable liquids spill away from exposing vessels with a BLEVE potential – drainage is not available when rainwater is being contained for environmental mitigations 10% or less of the time. This is in contrast to multiple jeopardy events, where two or more failures shall occur simultaneously (i.e. not in a compounded fashion) and such events are generally deemed incredible due to very low probability of occurrence.	1
Redundancy	For Event Scenarios where 100% reliable redundancy is available to prevent the initiating event, this CME would be appropriate. For example, two completely redundant, 100% capacity, automatic start-up instrument air compressors are available – if one fails, the backup redundant compressor would prevent the initiating event. For 200% totally independent redundancy, 2 CMEs are available. 300%, 3 CMEs, etc. Caution: All CME convention rules shall be met.	1 2(+) Ref. Section 3.1.3.1

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CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
Bull Plug Program	This applies to Event Scenarios such as “Vent or Drain Valve Left Open in Error”, or similarly a “Vent or Drain Valve Leaks By” where a bull plug would prevent the scenario from occurring. If the site has a program that assures that bull-plugs are installed and maintained at the end of open ended lines containing flammable or toxic materials, a CME would be appropriate. The KMEs for this CME would include a documented and audited program for routine monitoring, plug replacement, and LDAR detection (where applicable).	1
Robust Process Isolation Barriers	This applies to Event Scenarios involving a potential inadvertent valve operation that could result in personnel exposure to high pressure or corrosive/toxic/flammable material, or downstream equipment damage due to high pressure or other undesired factors. For the CME to be credited, there must be a program in place that requires a documented valve status verification for each isolation by a secondary personnel. The KME for this CME includes a documented and audited isolation/permitting program, such as the Integrated Safe System of Work (ISSoW), incorporating clear instructions for secondary personnel checks. This CME cannot be credited with other CMEs involving Human Intervention. In addition, the CME cannot be credited if the inadvertent valve operation is postulated to take place after the system has been verified by the secondary personnel.	1

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Toxic Vapor or Gas Release Event Scenarios - Prevent release or substantially reduce the Probability or Consequence of a release

CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
Emergency Shutdown Devices (ESDs), Burner Management System, and other Safety Instrumented Systems (SIS)	This CME group consists of sensor(s), logic processor(s) and final element(s) that detect out-of-limit conditions and bring the process to a functionally safe state without human intervention. See Section 3.2 of this Guideline for the Idd El Shargi Control Systems CMEs, should full SIS requirements not be met.	SIL 1 = 1 SIL 2 = 2 SIL 3 = 3 Ref. Section 3.1.3.1
Innately Safe Design	This CME may be credited if it can be demonstrated that the design prevents this Event Scenario or Significantly mitigates the consequences. This CME cannot be credited in addition to other CMEs, such as the Innately Safer Mechanical Integrity Design CME (PR-7) below, unless the two are totally independent. Examples include: 1) a pump whose maximum discharge pressure does not exceed the design pressure of the downstream equipment, and 2) valve actuation time that is sufficiently slow to avoid an overpressure situation. Note: A short comment in the PHR shall be provided for future reviewers to understand the credit and the factors that could otherwise allow the Event Scenario to be considered “incredible”.	1
Human Intervention	A CME can be assigned when one or more system(s) or device(s) requires human intervention to take preventative or mitigative action. This CME can be assigned ONLY IF all of the following are true:	1

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	1) The system or device is monitored 100% of the time by personnel, 2) Personnel have sufficient time to analyze the situation and take the required action that will prevent or Mitigate the Event Scenario, and 3) All necessary KME are in place, such as procedures, training, testing and drills. This CME can be credited in addition to an automatic control or ESD system CME if the Human Intervention system is totally independent¹ of these other systems. Continued on next page	
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CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
	Offshore Specific A CME can be credited for offshore operations where human intervention, in terms of escape, could mitigate the consequences. This CME can be if assigned ONLY IF in addition to the above, all of the following are also true: 1) Personnel working in the plant area have full knowledge of escape routes, 2) Carry personal gas detectors, and 3) Have immediate access to BA sets.	
Pressure Relief Devices (PRDs)	PRD's prevent systems from exceeding specified overpressures. For this CME to be applicable, potential consequences resulting from the PRD flow shall be evaluated (e.g., flammable gases released into congested areas). Effectiveness of PRDs is sensitive to service and experience. All necessary KMEs shall be in-place (e.g., testing, calibration, maintenance) to ensure the systems will not degrade to a level where failure is likely to occur.	1
Scrubbers and Flares	This CME group constitutes an integral part of a pressure relieving system that provides a level of protection by Mitigating the Consequences associated with a flammable gas release. This CME can only be applied when credit for those systems is not taken as an integral element in another CME such as ESD.	1
Detection and Control Systems (Toxic Gas, Flow, Pressure, Temperature)	This CME group consists of sensor(s), logic processor(s) and final element(s) that detect out-of-limit conditions and bring the process to a functionally safe state. For CME credit, the system shall be totally independent of other CME credits.	1

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CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
	Typically, this is a Basic Process Control System (BPCS) such as a DCS, PLC, or relay logic control. See Section 3.2 of this Guideline on the Idd El Shargi Control Systems CMEs for clarification.	
Innately Safer Mechanical Integrity Design	Use of a material that is known to not be susceptible to corrosion, erosion, or other failure mechanisms by the process. Failure could only result from slow degradation with ample time to detect degradation before failure (i.e., multiple inspection cycles). To apply this CME all necessary KMEs shall be in-place to ensure the systems will not degrade to a level that failure could occur. Examples of KMEs to maintain the system design parameters include, as appropriate, cathodic protection, chemical corrosion program, a robust MI program for monitoring and testing at appropriate frequency, and MOC program.	1
Secondary Containment Systems (Dikes, Berms)	These systems include passive devices that are designed, maintained and demonstrated to provide a level of protection that substantially reduces potential effects on people, the environment and PD/LOR.	1
Drainage – Directs Toxics to Safe Location	This CME is applicable if hydrocarbons are directed away from other equipment, occupied structures, etc. to the extent that impacts and potential effects on people, the environment and PD/LOR are substantially reduced.	1
Car-Sealed or Locked Valve	For Event Scenarios where prevention of an inadvertent opening/closing of a manual valve would prevent the scenario, this CME would be appropriate. A system shall be in place (documented list of car seals/locks and periodic checks) to verify that car-seals or locks are in place and valves are in the correct position to be credited as a CME. Car-seals shall be of substantial design so as not to be easily breakable (e.g., cables or chains). Inadvertent operation of valves that are not readily accessible or are so large as to be difficult to operate	1

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CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
	would typically be an incredible Event Scenario. This CME cannot be credited if the valve is the same valve for which credit is taken in IF 7.	
Sampling (Manual)	This CME is applicable if sufficient time exists to analyze multiple samples and take corrective actions so as to be considered reliable and effective. This CME cannot be credited with other CMEs involving Human Intervention.	1
Spare Equipment (Manual Startup or Activation)	This CME is applicable if sufficient time exists to bring the equipment onstream to prevent or mitigate the Event Scenario. The Event Scenario shall not involve a common cause failure (e.g., loss of power would affect both an operational and spare pump). This CME cannot be credited with other CMEs involving Human Intervention.	1
Spare Equipment (Automatic Activation)	This CME is applicable if the Event Scenario does not involve a common cause failure (e.g., loss of power would affect both an operational and spare pump).	1

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Toxic Vapor or Gas Release Event Scenarios - Mitigate People Consequences

CME Group	Description: These CMEs will be credited when they achieve the objective of their CME Group.	CME Credit
Layout/Spacing/ Separation Distance	This CME can be credited if layout/spacing/separation distance either prevents significant injuries or allows people enough time to escape without significant injuries OR if wind rose demonstrates that wind direction will prevent people consequences at least 90% of a year.	1
Process Isolation Valves	If the toxic inventory holdup within the process equipment and piping can be automatically isolated to substantially reduce the amount of toxics available for release, a CME will be credited. This CME credit can only be taken if it is totally independent ¹ of shutdown system, process controls, or human intervention that have already been claimed as CMEs.	1
Safe Shelter-In- Place	Occupied buildings protected by automatic detection, isolation and pressurization systems which protect occupants until a toxic cloud has passed or until immediately available emergency breathing air can be donned will be credited as a CME. To receive credit, reliable emergency routes shall also be available.	1
Contingent Element – Receptor Is Not Present > 90% of the Time	This CME will be credited for potential injury to personnel if personnel rarely (< 10% of the time) would be in the area affected by the event scenario. Examples include a process controls initiated excursion at a flare or a tank farm. If personnel are in such portions of a facility on a limited basis, then a CME credit can be applied to account for lower probability of personnel impact.	1
Preventative PPE Measures	This applies to an Event Scenario where an operator or maintenance person shall operate/maintain equipment such as a valve where his actions could potentially release toxics. A CME can be taken for positive pressure SCBA if the facility has a documented, audited program that assures that SCBA is used each time work is performed that could lead to this scenario. KMEs would include a respiratory protection program that complies with Idd El Shargi requirements.	1

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1. Independent systems for instrumented systems implies totally separate sensors, logic solvers, and final control element.

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APPENDIX B - GENERAL TREATMENT OF CONTROL SYSTEMS IN PHRS

PHR methodologies such as HAZOP and What-If consider control systems as both causes (initiating events) of process deviations that may have adverse consequences and as CMEs to prevent adverse consequences.

When Control System loop (or shutdown loop) failures are considered as causes of process deviations in Event Scenarios, the loop should be considered a “black box” in which any loop component can fail in a way that causes the final control element (typically a valve, switch, or similar devices that trip pumps, compressors, etc.) to respond in a manner that could lead to an undesired event. Using this convention, other discrete devices and components within the control loop (e.g., transmitters, indicators, recorders, control cards, valve stem, gate, etc.) are not analyzed for failure separately unless a loop component failure could result in a different consequence than arrived at using the black box approach.

As potential causes of process deviations, it should be assumed that actuated valves and other final control elements may fail in either a closed or open (or high and low) position irrespective of the design failure mode (e.g. a fail-close valve may actually fail in an open position). The basis for this PHR assumption is that various failure mechanisms may occur in a control loop as a result of process conditions, human errors in the design, maintenance, and operation which could affect the fail position. Though it is unlikely for a final control element to fail in a manner contrary to its design fail position, it is still a credible occurrence and, therefore, is considered in PHRs as a cause for process deviations. In some cases it may also be appropriate to consider a final control element failing in its last position as a cause for a process deviation such as during a shutdown mode of operation when the valve should fail closed.

CME credit cannot be taken for any device within a control loop that is assumed to fail (e.g., no credit would be given for a high pressure alarm that alerts an operator to take action if the alarm is part of the same loop for which the pressure control valve is assumed to fail in a certain mode). The basis for this assumption is that any number of devices within a loop may fail in a manner that prevents other components from being reliable

APPENDIX C - BASIS FOR CME CREDIT OF CONTROL SYSTEM LOOPS

CME Credit When Event Scenario is Caused by BPCS Loop Failure

During the development and evaluation of PHR Event Scenarios, BPCS loops are identified as causes (initiating events) of process deviations. For these scenarios, a conservative approach is to assume that no other BPCS loop will function as designed and, therefore, no other BPCS loops would be credited as CMEs to prevent adverse consequences. A less conservative and more reasonable approach as adopted by Idd El Shargi in this guideline is to allow one (1) other independent BPCS loop to be credited as a CME subject to specific criteria in Section 3.2.2. This approach is based on the following BPCS features:

- The BPCS logic solver, a common component to BPCS loops typically has a much lower probability of failure than other components.
- Logic solver failure is generally designed to bring the process to a fail-safe state.
- Many BPCS loops operate in a continuous mode of operation or are periodically switched with backup equipment, thereby reducing the likelihood

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that a valve will fail to function or that a switch will fail to activate (de-energize) when necessary.

- Many process systems have more than two (2) BPCS loops which are independent in terms of sensors and/or final control element.

Collectively, the above features are believed to provide the level of reliability expected from a CME and for the specific case of when a different BPCS loop failure initiates an Event Scenario.

CME Credit for Two (2) BPCS Loops

When multiple BPCS loops have been identified which would prevent or significantly mitigate an Event Scenario, only one (1) CME would normally be credited because the loops may lack independency to be highly reliable in terms of the following factors:

- Common BPCS logic solver
- Common input / output cards
- Non-redundant data communications
- Loss of utility (electricity, air, nitrogen, etc.)
- Programming errors
- Change / security control for programming and alarm bypassing

Multiple BPCS loops considered as CMEs would have independent sensors and final control elements, but would share the logic solver and may also share input / output cards. Following a strict rule of independency would mean that only one BPCS loop could be credited as a CME in an Event Scenario. In relation to Layers of Protection Analysis (LOPA), a CME is similar to an Independent Protection Layer (IPL). As described in the Centre for Chemical Process Safety (CCPS) book on LOPA, this would be a conservative approach (Approach A) in that credit for only one (1) IPL would be allowed. This approach has the benefit of being simple to evaluate in PHRs because once a BPCS loop is credited as a CME, then no others are considered.

The CCPS book on LOPA also allows for a second “CME” by following Approach B in which credit is given for up to two (2) independent loops even though they share a common logic solver. The basis for this approach is historical data that suggests the probability of a logic solver failure is much lower than the sensor and final control element within a loop. So, the overall probability of failure on demand (PFD) may not be appreciably greater than if there were two entirely independent loops. However, other factors mentioned above may reduce the independence and reliability of the BPCS loops, leading to this approach being non conservative. Therefore, other criteria has been developed in this guideline (Sections 3.2.3 – 3.2.4) that is consistent with LOPA Approach B to provide assurance that there is a high level of independence and reliability when two (2) BPCS loops are to be credited as CMEs. These criteria are considered very rigorous and would not be met unless the BPCS were of highly redundant design and were appropriately managed in terms of security and access control.

Consequently, in many Event Scenarios the crediting of two (2) BPCS loops as CMEs will not be possible.

Two (2) BPCS loops would typically provide a risk reduction in the range of about 100 and as much as 1,000 based on the Idd El Shargi Risk Matrix. This implies a probability of failure

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on demand (PFD) of 10^{-2} to 10^{-3} which is similar in performance to a SIS function that is rated to SIL 2. ANSI/ISA Standard 84.00.01-2004 (IEC 61511 Mod) Paragraph 9.4.3 conservatively does not allow a risk reduction factor of 10 or more for BPCS functions unless it is designed to the requirements of the standard.

Allowing two (2) BPCS loops to be credited as CMEs is based on:

- The rigorous criteria required for crediting two (2) BPCS loops as CMEs provides system characteristics similar to a SIL 2 system.
- Consistency with RAGAGEP (CCPS LOPA, Approach B).
- Noted that some companies use a PFD of 10^{-1} for each BPCS IPL which would equate to a risk reduction factor of 100 for two (2) BPCS loops.

If the Event Scenario identifies a BPCS loop as a cause for a process deviation, then only one other independent BPCS loop may be credited as a CME subject to the criteria of Section 3.2.2.

CME Credit for SIS and IPS Loops

SIS are designed to provide safety functions that are very reliable if maintained to the design SIL. A higher SIL equates to a lower Probability of Failure on Demand (PFD) of a dangerous failure by an order of magnitude as shown in the following table for demand mode of operation (typical).

DEMAND MODE OF OPERATION		
Safety Integrity Level (SIL)	Target average probability of failure on demand	Target Risk Reduction
4	$\geq 10^{-5}$ to $< 10^{-4}$	$> 10,000$ to $\leq 100,000$
3	$\geq 10^{-4}$ to $< 10^{-3}$	$> 1,000$ to $\leq 10,000$
2	$\geq 10^{-3}$ to $< 10^{-2}$	> 100 to $\leq 1,000$
1	$\geq 10^{-2}$ to $< 10^{-1}$	> 10 to ≤ 100

A CME provides a risk reduction factor between 10 and 100 which is the same as a SIS designed to SIL 1. Each increase in SIL provides a ten-fold risk reduction and so, mathematically, the risk reduction provided by an additional CME is higher than provided by a step increase in SIL. However, as there is a relatively close correlation between risk reduction afforded by CMEs and SIL, particularly for SIL 1 and 2, the number of CMEs is equated to the SIL for a SIS. Therefore,

SIL 1 = 1 CME

SIL 2 = 2 CMEs

SIL 3 = 3 CMEs

Note: A requirement for a SIL 3 SIS should be very rare in the oil and gas industry and SIL 4 should not apply as the complex design, maintenance, and extensive testing requirements would make such a system prohibitive in terms of cost and/or operation.

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A single SIS loop (e.g., high pressure shutdown) may be designed to SIL 2. Therefore, 2 CMEs may be credited for such a system.

PHR teams may consider failure of a SIS as an initiating event for an Event Scenario. By standard HAZOP convention, no other CMEs could be credited for other functions provided by the SIS because they would, in many cases, be part of the loop that has failed. For example, if an Event Scenario considered a spurious trip of an ESD valve as a cause for a process deviation, other related SIS functions such as alarms / human actions could not be credited as CMEs. As SIS are designed to be highly reliable, not allowing any credit for other SIS functions that may prevent an Event Scenario did not appear to be a rational approach and would lead to unreasonably higher risk levels. Therefore, CME IF-6 was created to recognize the fact that SIS failures are low frequency events.

The term IPS is used to define SIS that may have been designed to ANSI/ISA 84.00.01 (IEC 61511Mod) but not documented to have been operated or maintained to the extent required by these standards. IPS may also be shutdown systems that are distinct from the BPCS but were not constructed to one of these recognized standards. In effect, IPS are SIS for which the specified reliability cannot be sufficiently demonstrated and, therefore, these systems are treated similarly to BPCS in terms of crediting CMEs. If an IPS were to be updated, maintained, and tested, per all of the ANSI/ISA 84.00.01 (IEC 61511 Mod) requirements, then it could be re-established as a SIS and be credited for CMEs according to the SIL.

APPENDIX D – REVISION HISTORY LOG

TABLE 6: REVISION HISTORY LOG

Revision Number: 01

Date: 07/12/2023

Item Revised	Revision Description	Page No.
All	Document simplified as part of QMR initiative. No change to technical contents.	All
Remarks:		

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APPENDIX E - LIST OF ABBREVIATIONS

TABLE 7: LIST OF ABBREVIATIONS

Abbreviation	Description
CME	Critical Mitigation Element
PHR	Process Hazard Review
DCS	Distributed Control System
PLC	Programmable Logic Controller
NRO	Non Routine Operations
LTI	Long Term Inhibit
KME	Key Mitigations Element
BPCS	Basic Process Control System
SIS	Safety Instrumented System
SIF	Safety Instrumented Function
IPS	Instrumented Protective Functions
PFD	Probability of Failure on Demand
MTTF	Mean Time to Failure
TI	Test Intervals
MTTR	Mean Time to Repair
STR	Spurious Trip Rate
M oo N	M out of N
ESD	Emergency Shutdown System
BMS	Burner Management System